

Few-Shot Radar Annotation

Goals:

- Develop high-accuracy method(s) to detect, locate, and classify small and large objects in radar sweeps in any scenario.

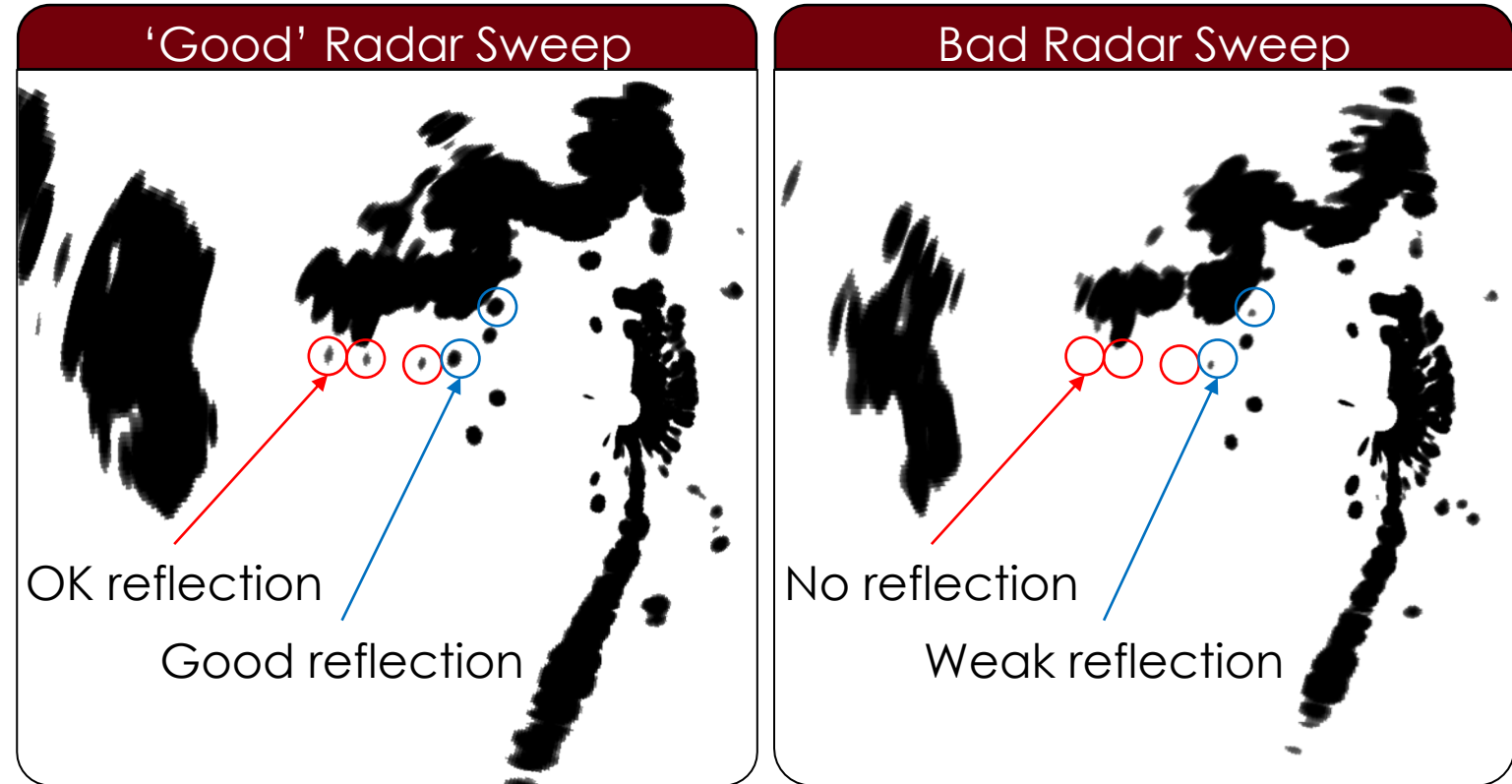
Progress Since Last QPR:

- Data capture trips across 9 field sites and 3 lakes, collecting over 100,000 sweeps
- Improved collection process; developed data storage method
- Explored HDR, radar characteristics, Standard tracking models, ML models, and Large Models (billions of params)
- Developed customizable non-electromagnetic simulations for radar sweeps

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Challenges in radar annotation methods

- Small objects are invisible to consumer-grade radar in sea clutter
- Consumer-grade radar operates with limited 4-bit dynamic range
- Radar gain requires constant readjustment

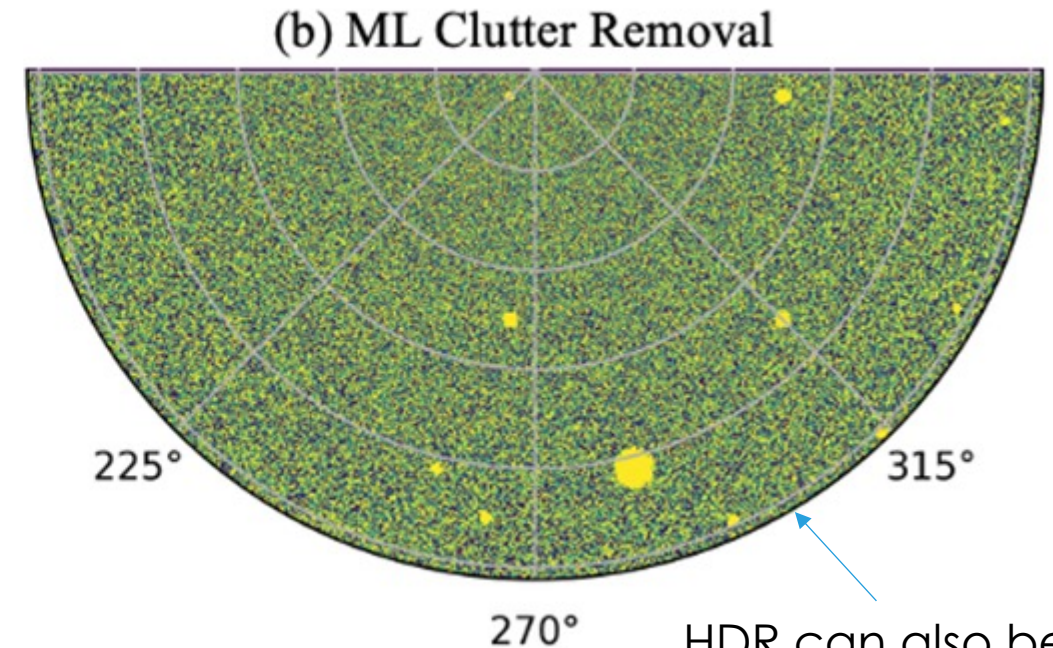
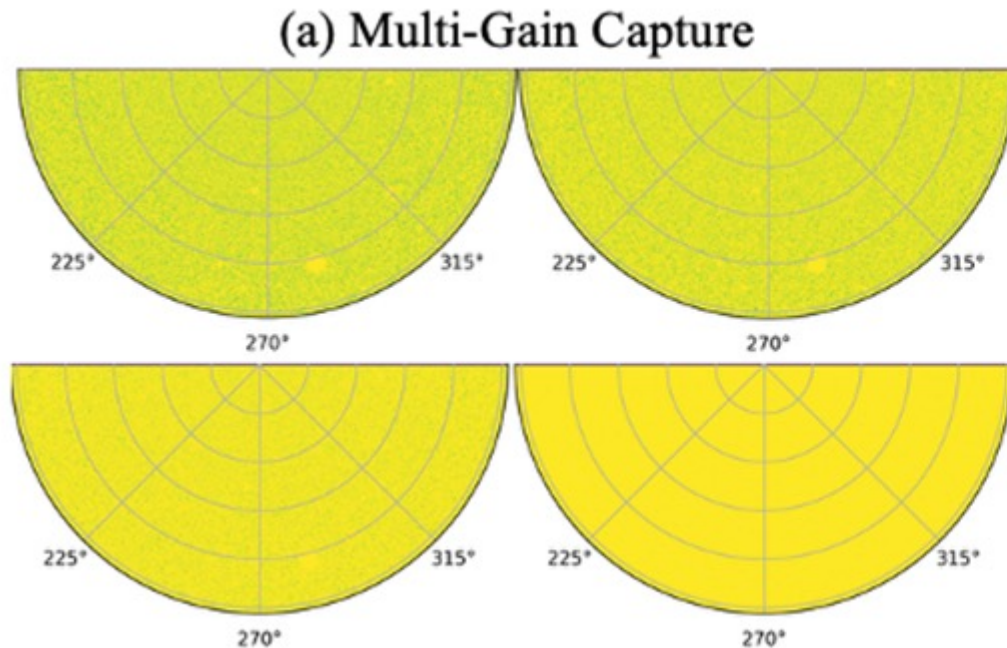


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Overview of HDR & clutter removal

Tossing 'cluttered' frames into a sufficiently intelligent algorithm, a clear picture can be extracted.

Apply this idea across time(temporal space), and object 'class' can be derived



HDR can also be applied at this stage

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Objectives and Results

Overview of Deliverables

- **Multi-Gain HDR Radar Software**
 - Combine multiple gain sweeps for extended dynamic range
- **Self-Supervised Learning**
 - Train detection models with minimal human annotation
- **Open Dataset**
 - Create an open dataset for maritime radar ML research

Engineering Targets

- **Classification & Detection:**
 - $>95\% P_d$ with $10^{-2} P_{fa}$
- **Labelling Target:**
 - <5 labeled frames per site for few-shot learning
- **Dataset Target:**
 - 200+ unique collection sites with 2 hrs of data at each under a variety of weather conditions

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Current Progress (May-August)

(MAY) Foundation

- Theoretical framework development
- Multi-gain concept introduction
- Initial hardware integration
- CFAR analysis and implementation

(JUNE-JULY) Data Collection



(AUGUST) ML Testing

- HDR system validation
- Tracking algorithms implemented
- Radar simulation development
- Corner reflector characterization

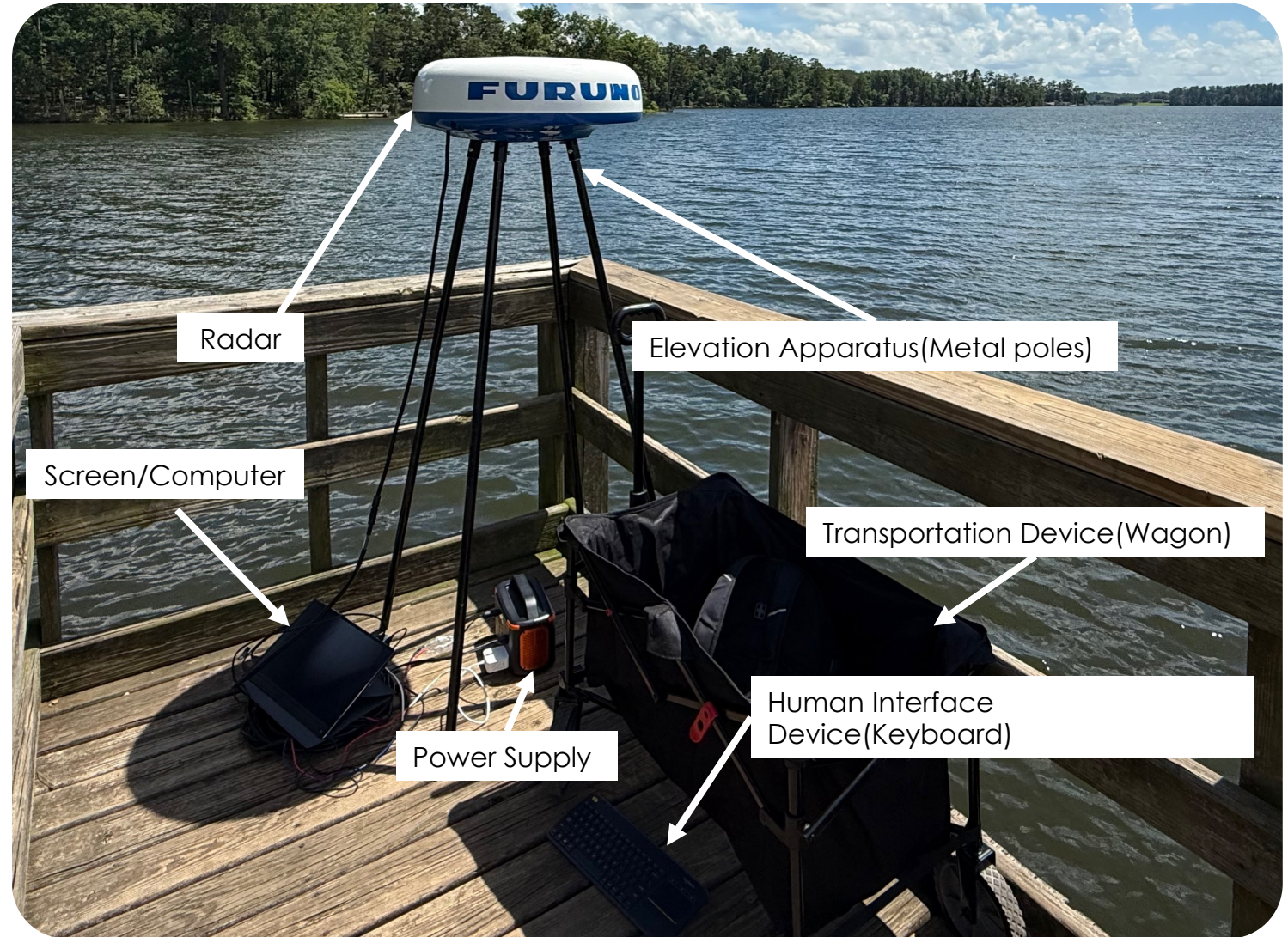
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Most Common Data Collection

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Collection Improvements

- 75% reduction in primary codebase (2000 to 500 lines)
- >90% reduction in deployment
- Boot time: 4-6 minutes to <20 seconds
 - (total setup <5 minutes)
- Enhanced reliability and portability



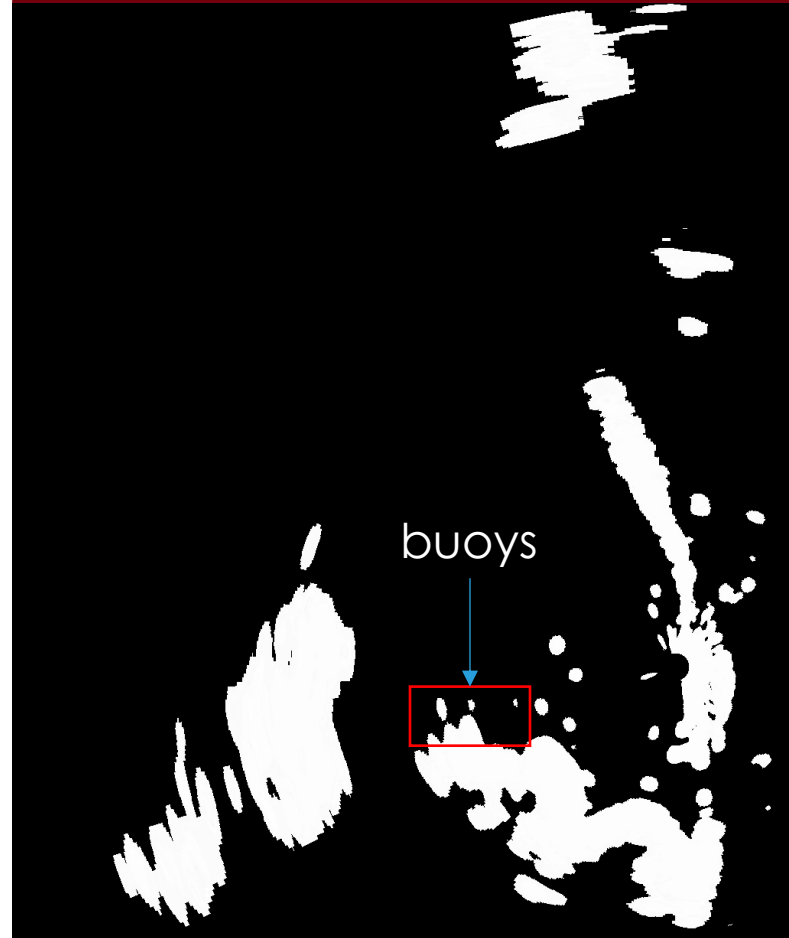
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High Dynamic Range Implementation

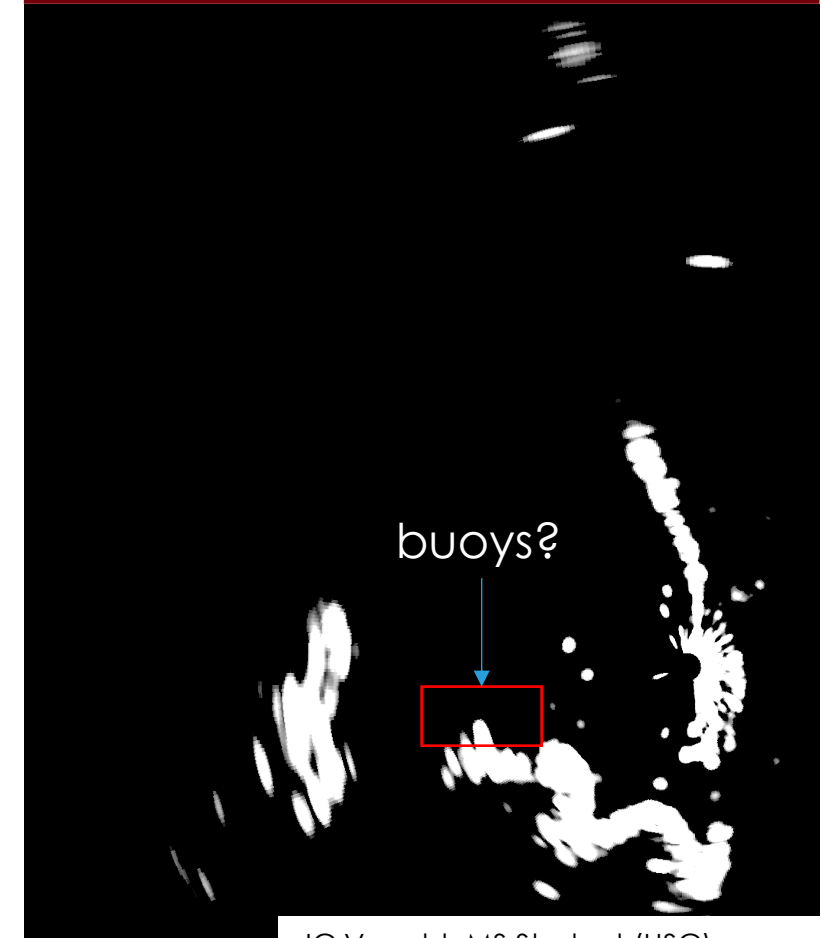
Analysis & Comparison

- **Robertson Method** (chosen)
Sigmoid response,
preserved faint & bright
echoes
- **Mertens Merge**: Good
performance, secondary
choice
- **Debevec**: Standard HDR,
moderate results
- **Stacking Method**: In-house
method, underperformed

HDR Sweep



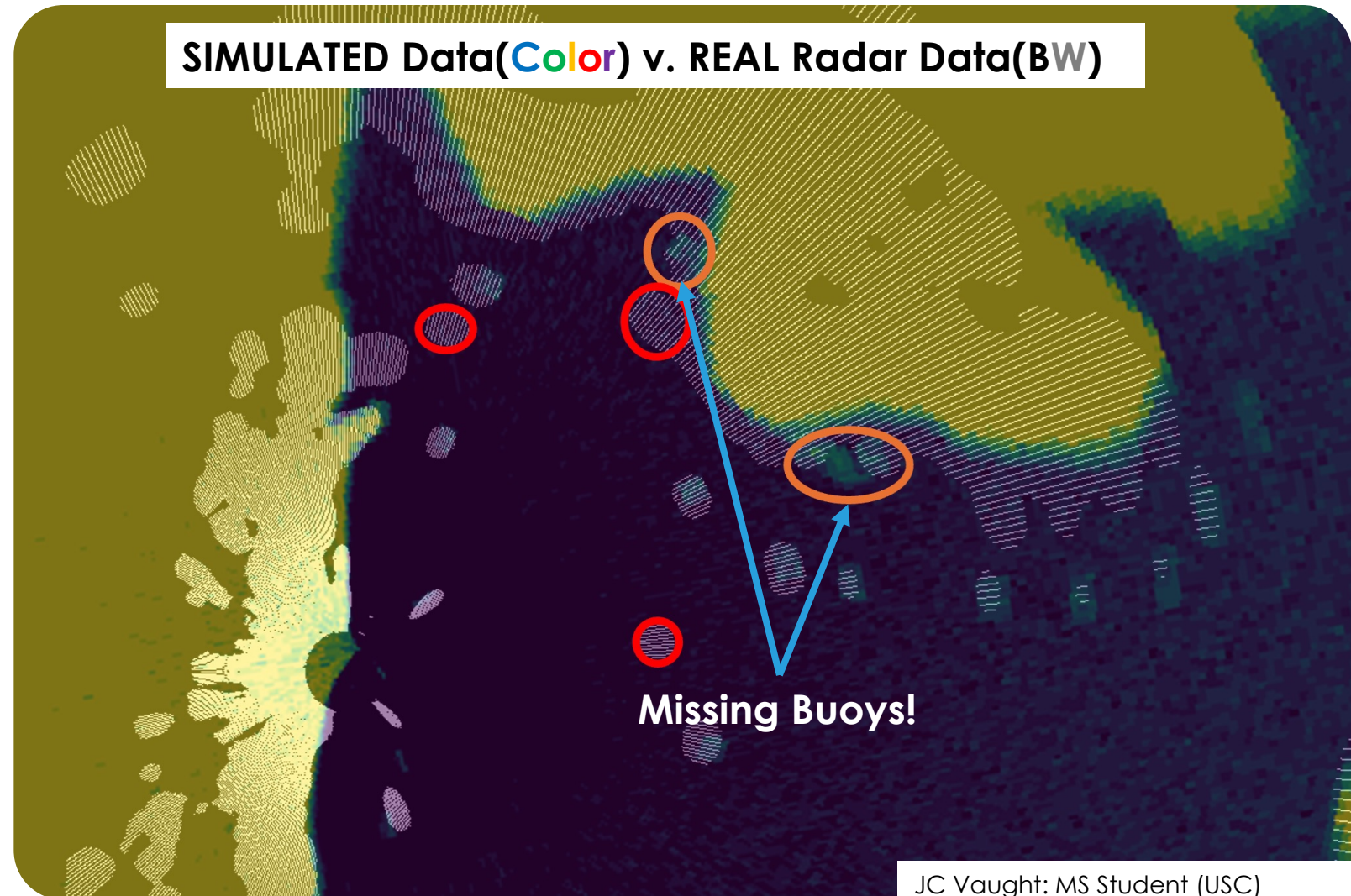
Single Sweep



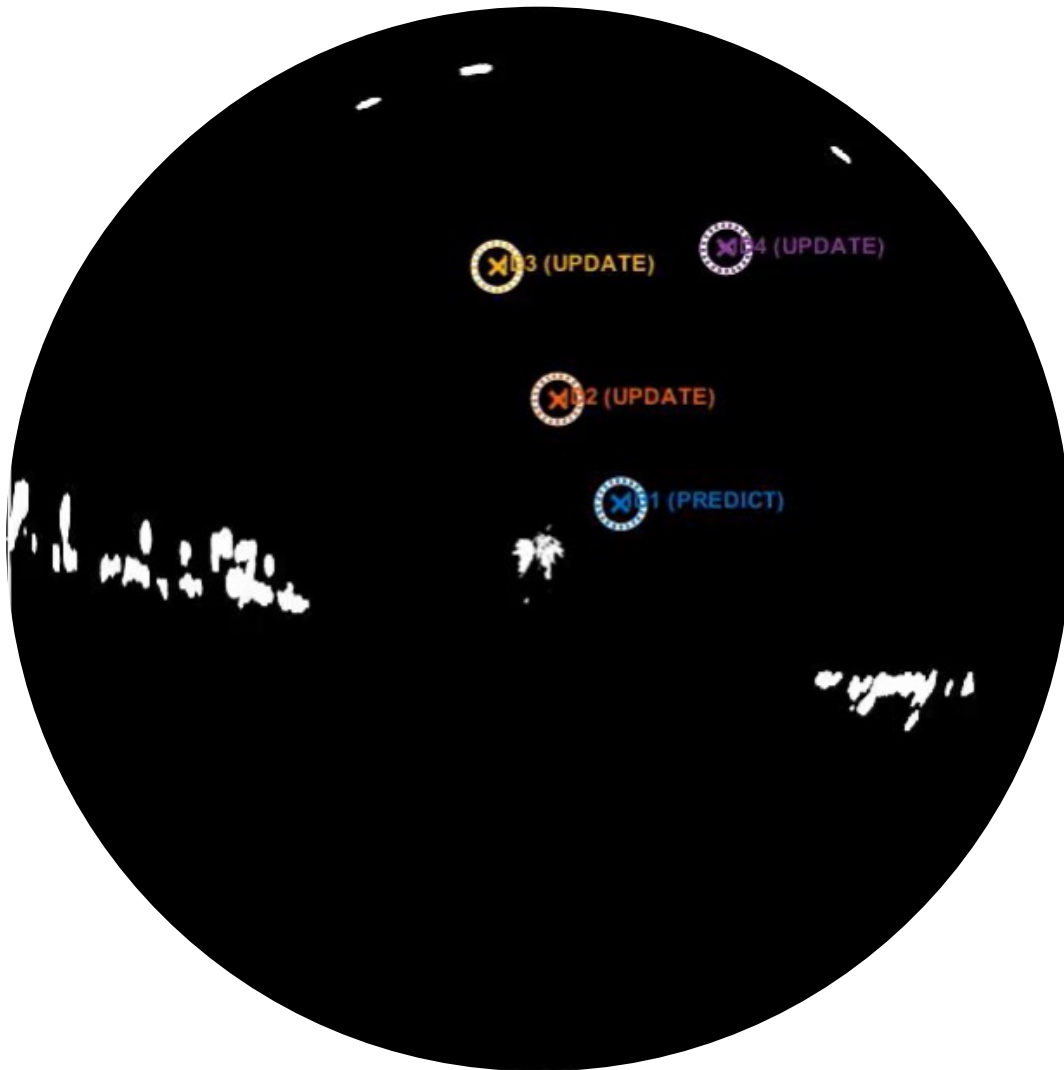
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Geographically Accurate Testing Environment

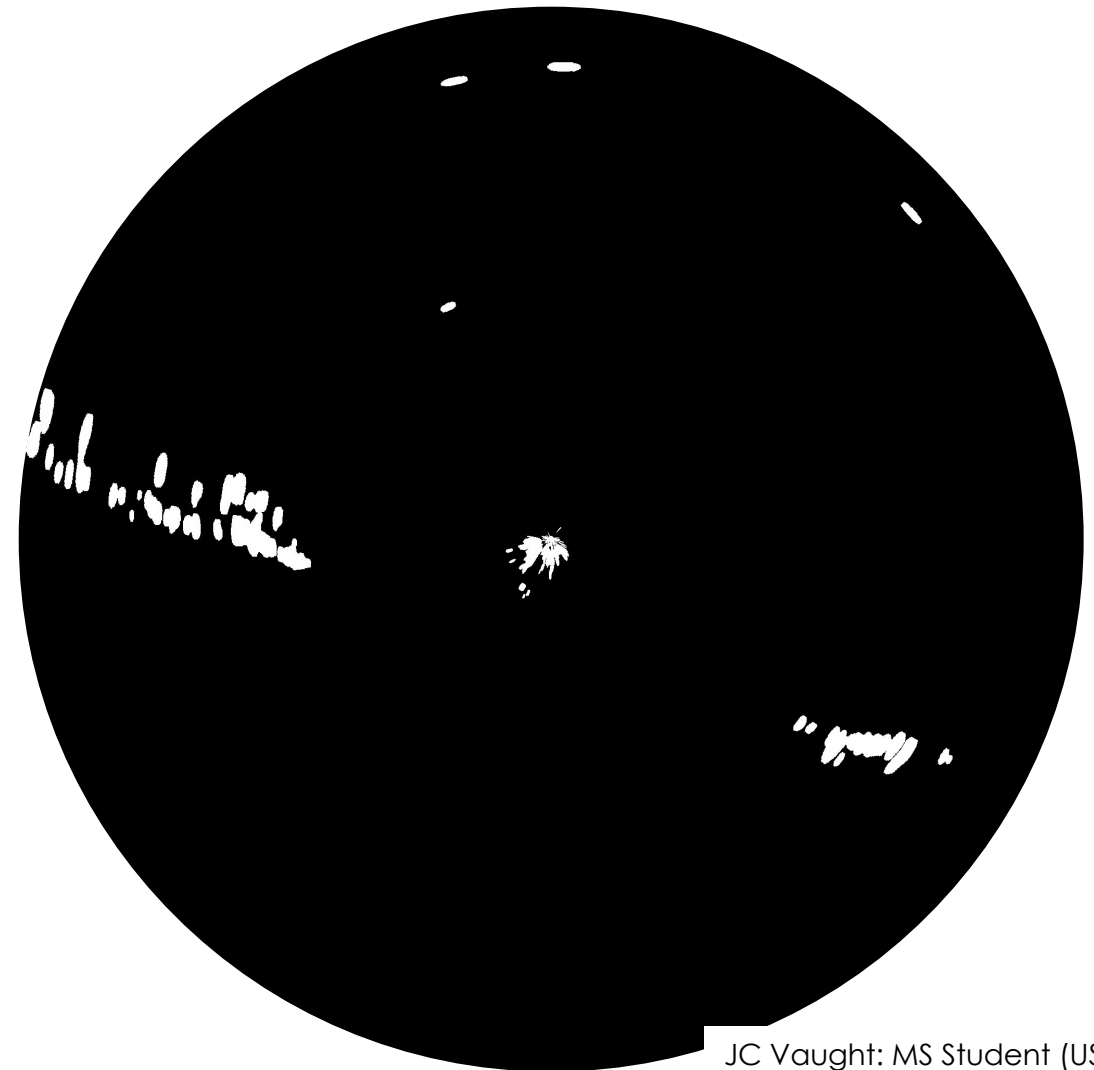
- K, log-normal, Weibull, Rayleigh distributions for clutter
- Automatic shoreline data from public sources
- Cell-by-cell power map generation
- External antenna pattern tables & STC curves
- Quantization matching radar front-end (low-bit)



Manual tracking



Automatic tracking



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Summary & Future Work

Completed Novel (=Publishable) Work

- Shannon entropy optimization replacing subjective gain adjustment
- HDR approach applicable to low-bit radar systems

Future Work/Currently In-Progress

- Geographically accurate radar simulation for controlled testing
- Large Model testing for masked auto-encoder(MAE) pipeline
- Multi-Pulse HDR system
- Temporal MAE for few-shot maritime object detection
- CNN + Kalman Object Tracking
- Few-Shot Radar Sweep Segmentation Models

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